Test Planning Document for TaskAnchor

Version: v1.11

Last Updated: 05/01/2026

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1. Purpose

This document defines how the TaskAnchor system is tested to ensure all MVP features function correctly and support continuous task follow-through.

The purpose of testing is to verify that implemented MVP behavior works as expected across backend logic, frontend UI behavior, integration points, browser flows, manual workflows, and practical usability of the existing MVP screens.

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2. Scope

This testing plan covers MVP features defined in the SRS and validation activities needed for the capstone.

In scope:

- Authentication (register, login)

- Angular UI (Register, Login, Task List, Task Create)

- Task management (create, edit, delete, view)
- Task status transitions
- DueDate and overdue logic
- PriorityLevel
- Task sorting
- Progress Log (add + view)
- NextAction
- Refresh signaling across UI
- Integration testing (UI + API)
- End-to-end testing (Playwright)
- Manual MVP Workflow Testing
- Practical UX Review for existing MVP flows only
- Practical structure/access review for existing MVP screens
- Cosmetic demo-readiness styling for existing MVP screens only
- Existing authentication screen clarity/access corrections

Out of scope:

- New MVP features
- New workflows
- UI polish beyond basic clarity and access
- Major UX redesign
- Performance testing
- Multi-user scenarios
- Future enhancements

- Subtasks
- Archiving/restoring
- Reactivation of completed tasks
- Categories/tags
- Notifications/reminders
- AI features
- Payments
- Advanced prioritization
- Navbar/logout/session guard behavior
- Completed-task restore or undo behavior

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3. Test Approach

Testing is performed using six layers:

1. Manual Testing

Focus:

- Full workflows in the running application
- UI + API behavior
- Confirmation that implemented MVP behavior works end-to-end
- Confirmation that existing-screen clarity/access corrections work in the browser

Manual testing verifies that the user can complete core TaskAnchor workflows without relying only on automated tests.

2. Backend Logic Testing (xUnit)

Focus:

- Overdue logic
- Sorting logic
- Status transitions
- LastUpdatedDate updates
- Completed task behavior
- Progress Log persistence behavior
- Authentication API validation behavior

Backend tests verify that core system rules remain correct and that derived behavior is not stored incorrectly.

3. Frontend Component Testing (Karma)

Focus:

- UI rendering
- Angular binding using ngModel
- Event handling
- Service calls

- State-driven UI behavior

- Refresh signaling

- Task object shape consistency

- Tested structural containers for cosmetic demo-readiness styling

- Existing-screen authentication access and error-message behavior

Strict rules:

- No direct DOM manipulation

- No document.querySelector usage in implementation

- State-driven validation only

- Returned/rendered frontend Task objects must use the full Task object shape

- Create payload tests may use create-payload-shaped objects because they test request payload behavior

- Cosmetic styling tests must remain limited to structure/access support for existing MVP screens only

- Authentication screen corrections must remain limited to existing Register/Login MVP behavior

Latest Angular Karma run:

- Date: 05/01/2026

- Command: ng test

- Result: 120 of 120 tests passed

- Failed: 0

- Browser: Chrome 147.0.0.0 on Windows 10

4. Integration and E2E Testing (Playwright)

Focus:

- UI to API interaction
- Full browser flows
- Navigation behavior
- Page load behavior
- Existing MVP workflows only

Key integration validations:

- refreshTasks() propagation
- UI updates after API success
- No manual shared-state mutation
- Login to Task List navigation
- Register route access
- Task creation flow
- Status update flow
- Progress Log flow

5. Manual MVP Workflow Testing

Manual MVP Workflow Testing verifies that implemented MVP behavior works end-to-end in the running application.

This is separate from automated test coverage and is used to confirm real user flow behavior.

Manual workflow testing is limited to existing MVP behavior only.

6. Practical UX Review

Practical UX Review verifies clarity, access, and usability of existing MVP screens.

This is not new feature development.

Allowed focus areas:

- Clear labels
- Understandable task flow
- Practical structure and access for existing MVP screens
- Visible task status
- Visible NextAction
- Visible Progress Log entries
- Basic form readability
- Demo-ready usability
- Cosmetic styling for existing MVP screens only when it supports clarity and demo-readiness
- Visible error messages for existing Register/Login failures
- Navigation links between existing Register and Login screens

Not included:

- New workflows
- New features

- Major redesign
- Visual polish beyond basic clarity and access
- Navbar/logout/session guard behavior
- Completed-task reactivation, undo, restore, or archive behavior

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4. Test Environment

The system will be tested in a local development environment.

- Frontend: Angular (TaskAnchor.UI)
- Backend: ASP.NET Core API
- Database: SQL Server
- ORM: Entity Framework Core

Tools:

- Swagger
- xUnit
- Angular Karma
- Playwright
- Manual browser testing

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5. Features to Test

- Register (UI + API)
- Login (UI + API)
- Register route access
- Login/Register navigation links
- Register success navigation back to Login
- Register duplicate-email error display
- Login invalid-credential error display
- Navigation to /tasks
- Create task
- Edit task (UI + API + refresh)
- Delete task (UI + API + refresh)
- View active tasks
- Status transitions (Active → InProgress → Completed)
- Completed task terminal behavior
- Completed task exclusion from active list
- DueDate behavior
- Overdue logic (derived)
- PriorityLevel handling
- Sorting rules
- Add Progress Log entry
- View Progress Logs

- NextAction display
- Refresh signaling behavior
- Practical UX of existing MVP flows
- Manual workflow completion for implemented MVP behavior
- Cosmetic demo-readiness styling for existing MVP screens only

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6. Test Cases

TC-01: Register User (API + UI)

Expected:

- User created
- Input captured correctly
- Registration request sent to API
- Successful registration returns user to Login
- Duplicate registration is rejected with visible error feedback

TC-02: Login User

Expected:

- Valid login succeeds
- Login request sent to API

- User navigates to /tasks after success

- Invalid login is rejected with visible error feedback

TC-03: Create Task

Expected:

- Task created

- API called

- refreshTasks triggered

- Task visible after refresh

TC-04: Edit Task

Expected:

- Edit mode activated

- Inline editing works

- Save triggers updateTask

- refreshTasks triggered

- Changes persist after refresh

TC-05: Delete Task

Expected:

- deleteTask called

- API called
- refreshTasks triggered
- Task removed after refresh

TC-06: Status Transition

Expected:

- Active transitions to InProgress
- InProgress transitions to Completed after confirmation
- Completion confirmation can be cancelled without calling updateTaskStatus
- Completed is terminal
- updateTaskStatus called only after valid status action
- refreshTasks triggered after accepted status update

TC-07: Overdue Logic

Expected:

- A task is overdue when:
  - Status is Active or InProgress
  - DueDate exists
  - DueDate is before the current date
- Completed tasks are not overdue
- Overdue is derived, not stored

TC-08: Sorting

Expected:

- Tasks sort by:

  1. Overdue tasks
  2. Due date ascending
  3. No due date → priority High → Low

TC-09: Progress Log Add

Expected:

- Progress Log input displays for selected task
- Input captures text
- addProgressLog(taskId, text) called
- refreshTasks triggered
- Input clears after save
- Progress Log mode exits after save

TC-10: Progress Log Display

Expected:

- Progress Logs display per task
- task.progressLogs[] is rendered
- log.text or log.entryText displays correctly

TC-11: NextAction

Expected:

- NextAction is captured during task creation
- NextAction displays when present
- Empty NextAction does not break rendering

TC-12: Refresh Signaling

Expected:

- Task list reloads after:
  - create
  - edit
  - delete
  - status update
  - progress log
- UI updates use refreshTasks()
- Components do not manually mutate shared state

TC-13: Completed Task Behavior

Expected:

- Completed tasks are excluded from active task list

- Completed tasks do not reactivate
- Completed status remains terminal
- User receives confirmation before moving a task to Completed
- Cancelling completion keeps the task active and does not call updateTaskStatus
- Accepting completion updates the task to Completed and refreshes the active list

TC-14: Validation

Expected:

- Duplicate registration rejected
- Invalid login rejected
- Required API validations still behave correctly
- Duplicate registration shows visible frontend error message
- Invalid login shows visible frontend error message

TC-15: UI Rendering

Expected:

- Login page loads
- Register page is routed through /register
- Register page is accessible from Login page
- Login page is accessible from Register page
- Task List page loads
- Inputs and buttons are visible

- Empty task list message displays when no active tasks exist
- Existing MVP screens use tested structural containers for demo-readiness styling

TC-16: Task Object Shape Consistency

Expected:

- Returned/rendered frontend Task objects use the full Task shape:
  - taskId
  - title
  - description
  - status
  - priorityLevel
  - dueDate
  - nextAction
  - lastUpdatedDate
- Optional progressLogs may be included:
  - progressLogs?: ProgressLog[]
- Backend-only userId is not required in frontend Task fixtures
- Create payload tests may remain create-payload shaped

TC-17: Manual MVP Workflow Testing

Expected:

- Register user

- Log in
- Navigate to Task List
- Create task with title, priorityLevel, dueDate, and nextAction
- Confirm created task appears in Task List
- Edit task title
- Confirm edited task refreshes and displays updated title
- Change status from Active to InProgress
- Change status from InProgress to Completed
- Confirm completed task is removed from active list
- Add Progress Log entry to a task
- Confirm Progress Log entry displays after refresh
- Delete task
- Confirm deleted task is removed after refresh

TC-18: Practical UX Review

Expected:

- Login flow is understandable
- Register flow is understandable
- Task List is readable
- Existing MVP screen structure supports task follow-through
- Existing MVP screen access is understandable
- Create Task form labels are clear
- Status change interaction is understandable

- NextAction display is visible when present
- Progress Log entry flow is understandable
- Delete action is visible and predictable
- Empty task list message is clear
- Error messages are visible for failed Register/Login attempts

TC-19: Cosmetic Demo-Readiness Styling

Expected:

- Login screen renders inside tested .login-card structure
- Register screen renders inside tested .register-card structure
- Create Task form renders inside tested .task-form-card structure
- Task List content renders inside tested .task-list-card structure
- Styling remains cosmetic and does not add new workflows or features
- Existing MVP controls remain visible and usable
- Login, Register, Create Task, and Task List visual styling remains consistent
- Angular Karma regression passes after styling updates

TC-20: Existing Authentication Access Corrections

Expected:

- /register route uses RegisterComponent
- Login page renders a Register link
- Register page renders a Login link

- Successful registration navigates back to Login
- Duplicate registration displays backend-provided error text when available
- Failed login displays a clear Login failed message
- Angular tests cover these behaviors

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7. Current Automated Test Coverage

Backend:

- Rule coverage for overdue behavior
- Rule coverage for sorting behavior
- Rule coverage for status transitions
- API behavior coverage for implemented backend flows
- Authentication validation behavior for duplicate registration and invalid login

Latest backend xUnit run:

- Date: 04/29/2026
- Command: dotnet test
- Result: 25 of 25 tests passed
- Failed: 0
- Skipped: 0
- Build: succeeded

Frontend:

- RegisterComponent
- LoginComponent
- TaskListComponent
- TaskFormComponent
- AuthService
- TaskService
- Routing spec
- AppComponent

Frontend coverage includes:

- UI rendering
- Angular binding behavior
- Service interaction
- Status transitions
- Edit/save flow
- Delete behavior
- Progress Log full flow
- Refresh signaling
- Task object shape normalization
- Tested structural containers for Login, Register, Create Task, and Task List demo-readiness styling
- Register route exposure
- Login/Register navigation links
- Register success navigation back to Login

- Register failed/duplicate-email error display

- Login failed-attempt error display

Latest frontend Karma run:

- Date: 05/01/2026

- Command: ng test

- Result: 120 of 120 tests passed

- Failed: 0

- Browser: Chrome 147.0.0.0 on Windows 10

E2E:

- Login page loads

- Task List page load coverage started

- Register route is now exposed and should be eligible for future Playwright coverage

Current Playwright result:

- Full run: 3 passed, 6 skipped across Chromium, Firefox, and WebKit

- Filtered active run excluding skipped Register test: 3 passed

Current E2E coverage is limited to routed page-load smoke tests. MVP feature behavior is currently covered primarily by backend xUnit tests, Angular Karma unit/component tests, and manual browser verification. Previously skipped Register route coverage should be revisited because /register is now routed.

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8. Manual MVP Workflow Checklist

- [x] Register user
- [x] Log in
- [x] Navigate to Task List
- [x] Create task with title, priorityLevel, dueDate, and nextAction
- [x] Confirm created task appears in Task List
- [x] Edit task title
- [x] Confirm edited task refreshes and displays updated title
- [x] Change status from Active to InProgress
- [x] Change status from InProgress to Completed
- [x] Confirm completed task is removed from active list
- [x] Add Progress Log entry to a task
- [x] Confirm Progress Log entry displays after refresh
- [x] Delete task
- [x] Confirm deleted task is removed after refresh

Additional authentication/access manual confirmations:

- [x] Confirm Login page provides access to Register
- [x] Confirm Register page provides access back to Login
- [x] Confirm successful registration returns to Login
- [x] Confirm duplicate registration displays visible error message
- [x] Confirm invalid login displays visible error message

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9. Practical UX Review Checklist

- [ ] Confirm Login flow is understandable
- [ ] Confirm Register flow is understandable
- [ ] Confirm Task List is readable
- [ ] Confirm existing MVP screen structure supports task follow-through
- [ ] Confirm existing MVP screen access is understandable
- [ ] Confirm Create Task form labels are clear
- [ ] Confirm Status change interaction is understandable
- [ ] Confirm NextAction display is visible when present
- [ ] Confirm Progress Log entry flow is understandable
- [ ] Confirm Delete action is visible and predictable
- [ ] Confirm empty task list message is clear

Practical UX Review observation:

Initial manual screen review found that existing MVP screens are functional and core information is visible, including task status, priority, NextAction, Progress Log entries, and task actions. However, the initial layout and visual structure were plain and crowded.

A cosmetic styling/demo-readiness pass was completed for existing MVP screens only. Login, Register, Create Task, and Task List now use tested structural card containers with consistent black, white, and gold styling. The styling work remained cosmetic and did not add new MVP features or workflows.

Additional Practical UX Review/access observations:

- Login originally had no visible Register access.

- Register was implemented but not routed.

- Register success did not return the user to Login.

- Invalid Login attempts displayed no visible message.

- Duplicate Register attempts displayed no visible message even though the backend returned a duplicate-email error.

- Login/Register navigation links originally had inconsistent styling.

- These issues were corrected as clarity/access fixes for existing MVP authentication screens.

Practical UX Review remains pending until the full checklist is explicitly confirmed through manual screen review.

Practical UX Review Findings from Task List Screen Review:

- NextAction displays without a label.

  - Current problem: NextAction text appears as plain text, which makes it harder to distinguish from description or Progress Log content.

  - Planned fix: Display NextAction with a clear label, such as "NextAction:".

  - Scope: In scope as a clarity/access fix for an existing MVP screen.

- Progress Log displays without a label.

  - Current problem: Progress Log text appears as plain text, which makes it harder to distinguish from NextAction or description content.

  - Planned fix: Display Progress Log entries with a clear label, such as "Progress Log:".

  - Scope: In scope as a clarity/access fix for an existing MVP screen.

- Task cards may become cluttered when many Progress Log entries exist.

  - Current problem: Showing all Progress Log entries by default can make the task list hard to scan.

  - Possible future fix: Show the latest Progress Log entry by default and provide a small "View all Progress Logs" / "Hide Progress Logs" toggle.

  - Scope: Potentially in scope only if kept display-only and limited to existing Progress Log data.

  - Do not implement before the label clarity fixes.

- NextAction cannot currently be edited after task creation.

  - Current problem: NextAction is captured on create and displayed in the task list, but the current edit flow appears limited.

  - Scope: Possible existing-field edit gap, but larger than a label fix.

  - Do not implement before inspecting current edit tests and template behavior.

- Important task data hierarchy is weak.

  - Current problem: Title, priority, NextAction, and Progress Log content visually run together.

  - Possible future fix: Small structure/readability improvements only.

  - Scope: In scope only as Practical UX Review clarity/access for existing MVP screens, not as redesign.

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10. E2E Coverage Gaps

Current Playwright coverage:

- Login page loads

- TaskList page load test added

- Register route is now exposed and the skipped Register page load test should be revisited

Latest Playwright runs:

- Full run: 3 passed, 6 skipped across Chromium, Firefox, and WebKit

- Filtered active run excluding skipped Register test: 3 passed

Remaining E2E coverage gaps:

- Register page load test should be unskipped/revalidated now that /register exists

- Login → TaskList navigation test

- Task creation flow test

- Status update flow test

- Progress Log flow test

E2E expansion must stay limited to existing MVP flows only.

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11. Risks and Problem Areas

- Test object shape inconsistency

- Frontend/backend contract mismatches

- Refresh pattern misuse

- Overdue must remain derived

- Risk of Angular overengineering

- Limited authentication enforcement

- Integration issues not caught in unit tests

- E2E coverage currently limited to smoke tests

- Practical UX issues may block demo clarity even if automated tests pass

- Manual workflow issues may reveal behavior gaps not caught by unit tests

- Cosmetic styling must remain limited to existing MVP screens and must not become feature expansion

- Login/Register access and error feedback must remain clarity/access corrections, not expanded authentication/session management

- Completing a task is terminal and removes it from the active list; accidental completion risk exists

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12. Known Issues / Required Fixes

Backend Test Build Blocker - Resolved

A backend test build blocker occurred because `Microsoft.AspNetCore.OpenApi` was using version `10.0.5` while the API targets `net9.0`.

Resolution:

- Changed `Microsoft.AspNetCore.OpenApi` from `10.0.5` to `9.0.0`

- Re-ran `dotnet test`

- Result: 25 of 25 backend tests passed

Remaining note:

- Backend test run completed with 2 nullable-reference warnings in `ProgressLogTests.cs`.

Test Integrity

Task object shape normalization has been completed for checked frontend returned/rendered Task tests.

Required returned/rendered frontend Task structure:

- taskId
- title
- description
- status
- priorityLevel
- dueDate
- nextAction
- lastUpdatedDate

Optional:

- progressLogs?: ProgressLog[]

Create payload tests should not be forced into the full Task shape.

Valid create payload shape:

- title

- priorityLevel

- dueDate

- nextAction

E2E Coverage

Playwright coverage has started and remains limited to existing MVP flows only.

Completed/current E2E coverage:

- Login page loads

- TaskList page load test added

Previously skipped/current gap:

- Register page load test was skipped because RegisterComponent was implemented and unit-tested but no /register route was exposed.

- /register is now routed, so the skipped Register Playwright test should be revisited.

Required future E2E targets:

- Register page load route test

- Login → TaskList navigation test

- Task creation flow test

- Status update flow test

- Progress Log flow test

Manual Validation

Manual MVP Workflow Testing is complete for the currently confirmed checklist items. Browser verification confirmed Login, Register, Task List, Create Task, Edit Task, status transitions, Completed task exclusion, NextAction display, Progress Log save/display after refresh, Delete Task, deleted task removal after refresh, and refresh behavior.

Additional browser verification confirmed:

- Login page provides access to Register
- Register page provides access back to Login
- Register success returns to Login
- Duplicate Register attempt displays a visible error message
- Invalid Login attempt displays a visible error message
- Login/Register link styling is visually consistent

Practical UX Review remains pending until the full checklist is explicitly confirmed through manual screen review.

Cosmetic Demo-Readiness Styling

Cosmetic demo-readiness styling has been applied to existing MVP screens only.

Completed styling work:

- Login screen wrapped in tested .login-card structure
- Register screen wrapped in tested .register-card structure
- Create Task form wrapped in tested .task-form-card structure

- Task List content wrapped in tested .task-list-card structure

- Consistent black, white, and gold styling applied

- Font consistency applied across Login, Register, Create Task, and Task List cards

- Login/Register link styling aligned

- Angular Karma regression passed after styling updates: 117 of 117 tests passed

This work did not add new MVP features, new workflows, priority-edit behavior, redesign, or scope expansion.

Authentication Access/Error Feedback Corrections

Resolved:

- /register route was missing.

- Login page did not expose a Register link.

- Register page did not expose a Login link.

- Register did not navigate back to Login after successful registration.

- Invalid Login attempt displayed no visible message.

- Duplicate Register attempt displayed no visible message.

- Login/Register link styling was inconsistent.

Resolution:

- Added /register route to existing RegisterComponent.

- Added tested Register link on Login page.

- Added tested Login link on Register page.

- Added tested Register success navigation back to Login.

- Added tested Login failure message.

- Added tested Register failure/duplicate-email message.

- Displayed backend-provided duplicate-email text from `HttpErrorResponse.error`.

- Aligned Login/Register link styling.

Completed Status Accidental Click Risk - Resolved

Observation:

- InProgress tasks remain in the active list.

- Completed tasks are excluded from the active list.

- Completed is terminal and cannot be reactivated in the MVP.

- User identified that accidentally completing a task could remove it from the active list.

Resolution:

- Added a confirmation prompt before moving a task to Completed.

- Cancelling the confirmation keeps the task active and does not call updateTaskStatus.

- Accepting the confirmation updates the task to Completed and triggers refreshTasks.

- Active → InProgress behavior remains unchanged.

- InProgress → Completed behavior remains valid only after confirmation.

Scope boundary maintained:

- Did not add undo, restore, completed-task reactivation, archive, completed-task list, or a new status workflow.

- Completed-task reactivation remains out of scope.

Verification:

- Added frontend test coverage for cancelled completion confirmation.

- Added frontend test coverage for accepted completion confirmation.

- Updated existing InProgress → Completed test to confirm before completion.

- Angular Karma passed 120 of 120 tests.

- Manual browser verification confirmed Cancel keeps the task active and OK completes the task and removes it from the active list.

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13. Changelog

v1.11 (05/01/2026)

- Updated latest Angular Karma result to 120 of 120 tests passed using ng test.

- Added tested completion confirmation behavior before marking a task Completed.

- Added coverage for cancelled completion confirmation to verify updateTaskStatus is not called.

- Added coverage for accepted completion confirmation to verify updateTaskStatus and refreshTasks are called.

- Updated existing InProgress to Completed status test to account for confirmation behavior.

- Manually verified that cancelling completion keeps the task active.

- Manually verified that accepting completion completes the task and removes it from the active list.

- Resolved Completed Status Accidental Click Risk without adding undo, restore, reactivation, archive, completed-task list, or a new workflow.

v1.10 (04/30/2026)

- Updated latest Angular Karma result to 117 of 117 tests passed using ng test watch-mode.

- Added /register route coverage using red/green TDD.

- Exposed existing RegisterComponent through /register.

- Added tested Register link on Login page for users without an account.

- Added tested Login link on Register page for users who already have an account.

- Added Register success navigation back to Login using red/green TDD.

- Added tested .register-card structure for Register cosmetic demo-readiness styling.

- Applied Register black/white/gold styling aligned with Login, Create Task, and Task List.

- Added visible Login failure message using red/green TDD.

- Added visible Register failure/duplicate-email message using red/green TDD.

- Confirmed duplicate Register displays backend-provided message from HttpErrorResponse.error.

- Aligned Login/Register navigation link styling.

- Updated Manual MVP Workflow Checklist to mark Register user complete after manual browser verification.

- Added authentication/access manual confirmations for Login-to-Register access, Register-to-Login access, Register success return to Login, duplicate Register error display, and invalid Login error display.

- Updated TC-01 to include Register success navigation and duplicate-registration feedback.

- Updated TC-02 to include invalid-login feedback.

- Updated TC-14 to include visible frontend validation/error feedback.

- Updated TC-15 to include /register routing and Login/Register navigation links.

- Updated TC-18 to include Register flow and error-message clarity.

- Updated TC-19 to include .register-card styling.

- Added TC-20 for existing authentication access corrections.

- Updated automated frontend coverage notes to include Register routing, Register/Login links, Register success navigation, Login error display, and Register error display.

- Updated E2E notes because /register is now exposed and the skipped Register Playwright test should be revisited.

- Added known issue for accidental Completed status risk.

- Clarified that confirmation before Completed is allowed, but undo/restore/reactivation/archive/completed-list behavior remains out of scope.

- Kept Practical UX Review checklist unchecked pending explicit full manual screen review confirmation.

v1.9 (04/30/2026)

- Updated latest Angular Karma result to 111 of 111 tests passed using ng test --watch=false.

- Corrected frontend test count after removal of a duplicated Task List list-container render test.

- Added tested Task List structure for cosmetic demo-readiness styling using .task-list-card.

- Recorded red/green TDD flow for Task List styling support: structure test failed before wrapper, passed after wrapper.

- Added Task List cosmetic styling with black, white, and gold visual direction.

- Added font consistency across Login, Create Task, and Task List card containers.

- Documented that Login, Create Task, and Task List visual styling are now consistent.

- Added TC-19 for cosmetic demo-readiness styling coverage.

- Updated Practical UX Review observation to note that the initial plain/crowded layout was addressed through cosmetic styling for existing MVP screens only.

- Kept Practical UX Review checklist unchecked pending explicit full manual screen review confirmation.

- Updated Manual MVP Workflow Checklist to reflect confirmed browser verification for Login, Task List, Create Task, Edit Task, status transitions, Completed exclusion, NextAction display, Progress Log save/display after refresh, Delete Task, deleted task removal after refresh, and refresh behavior.

- Left Register user unchecked because that checklist item has not been separately confirmed in the current manual testing record.

v1.8 (04/29/2026)

- Recorded Manual MVP Workflow Testing confirmations for Progress Log display after refresh and Delete Task removal after refresh

- Updated latest Angular Karma result: 108 of 108 tests passed using ng test --watch=false

- Updated latest backend xUnit result: 25 of 25 tests passed using dotnet test

- Documented Manual MVP Workflow Testing as partially complete, with Practical UX Review still pending unless separately confirmed

- Added tested Login screen structure for demo-readiness styling; Angular Karma passed 109/109 after Login styling update.

- Added tested Login and Create Task structure for cosmetic demo-readiness styling; Angular Karma passed 110/110 after styling updates.

v1.7 (04/27/2026)

- Added Manual MVP Workflow Testing section and checklist

- Added Practical UX Review section and checklist for existing MVP flows

- Clarified that UX review is limited to clarity, access, usability, and structure of existing MVP screens

- Clarified that UX polish, redesign, new workflows, and new features are out of scope

- Added E2E Coverage Gaps section

- Updated test approach from four layers to six layers

- Updated known issues to reflect completed Task object shape normalization and remaining E2E/manual validation work

- Updated Playwright coverage notes to record latest run result: 3 passed and 6 skipped across Chromium, Firefox, and WebKit

- Documented skipped Register page E2E test as a routing/coverage gap because RegisterComponent is implemented and unit-tested but no /register route is currently exposed

- Marked TaskList page load Playwright coverage as started while keeping remaining E2E flows listed as gaps

- Recorded filtered Playwright run confirming 3 active E2E smoke tests passed after excluding skipped Register route coverage

- Recorded latest Angular Karma test run: 108 of 108 frontend tests passed using ng test --watch=false

- Resolved backend test build blocker by aligning Microsoft.AspNetCore.OpenApi package version with net9.0 target framework.

- Recorded latest backend xUnit test run: 25 of 25 tests passed with 2 nullable-reference warnings.

v1.6 (04/25/2026)

- Updated test coverage to reflect full frontend MVP completion

- Added edit flow testing (UI + API + refresh)

- Added full Progress Log testing (input, save, display)

- Added validation for refreshTasks across all operations

- Removed outdated gaps (task refresh, missing UI features)

- Introduced test consistency requirement for Task object shape

v1.5 (04/23/2026)

- Added refresh signaling tests

- Updated create-to-visible behavior

- Introduced ngModel binding validation

v1.4 (04/22/2026)

- Added Login UI, Task List UI, and Task Create UI to scope

- Added task creation behavior and service interaction testing

- Added login → task list navigation validation

- Documented known gap: task list does not refresh after creation
- Removed duplicate test case
- Updated frontend testing approach to include service interaction

v1.3 (04/20/2026)

- Added Integration and End-to-End Testing approach
- Added Playwright to testing environment
- Expanded scope to include full system testing (UI + API)
- Added TC-01B (Register page browser test)
- Added TC-01C (Register integration flow)
- Added TC-21 and TC-22 for Playwright-based E2E testing
- Updated risks to include integration and browser testing concerns
- Shifted testing emphasis from heavy TDD to full system validation per instructor feedback

v1.2 (04/18/2026)

- Added frontend testing approach section
- Added Register UI test case (TC-01A)
- Updated test environment to include Angular Karma
- Updated scope to reflect frontend implementation
- Updated risks to include frontend integration and complexity concerns

v1.1 (04/16/2026)

- Updated test cases to reflect implemented authentication endpoints
- Added tests for invalid login and duplicate registration handling

- Added test for Progress Log updating LastUpdatedDate
- Added test for correct task update using TaskId
- Updated test environment to reflect current frontend status
- Updated risks to reflect current implementation state